MIL-S-21294B(SHIPS)
6 February 1964
SUPERSEDING
MIL-S-21294A(SHIPS)
2 September 1960
(See 6.2)

MILITARY SPECIFICATION

SONAR SYSTEMS; CABLE, TERMINAL, AND WIRE TABLES FOR

1. SCOPE

1.1 This specification covers the compilation and arrangement of information, and the construction and arrangement of sheets comprising an installation booklet plan for a sonar equipment or system.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitations for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY
MIL-S-23140 - Drawing, Installation Control and Preliminary Data (for Electronic and Related Equipment).

STANDARDS

MILITARY
MIL-STD-2 - Engineering Drawings, Sizes and Formats.

PUBLICATIONS

MILITARY
DD-441 - Security Agreement and its Attachment, Industrial Security Manual for Safeguarding Classified Information.

(Copies of specifications, standards, drawings, and publications, required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements.- .

3.1.1 The installation booklet plan, conforming to the requirements specified herein, shall apply to all sonar equipment and system installations.

3.1.2 The booklet plan shall specifically meet all of the requirements specified in MIL-STD-2 and MIL-D-23140 except in horizontal dimensions.

3.1.3 The booklet plan shall be constructed of size A sheets and shall be finished in a neat and workmanlike manner.

3.1.4 Material.- The material for the Installation Plan Booklet permanent record sheets for masters shall be right reading, ink, cloth tracings or black line cloth photo-tracings. Permanent record sheets shall be neat, clean and free from smears, with lines and characters that are sharp, clear, durable, undistorted, and provide good contrast to the background. Tracings shall be capable of producing distinct direct copy prints to the fourth generation.

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3.1.4.1 Substitution.- The Bureau of Ships shall decide the applicability for use of the material proposed by the contractor. If the contractor is in doubt about the type of material or process, or has some new method which could save time or cost, a sample shall be furnished to the Bureau of Ships, Code 679C4, for approval.

3.2 Contents.- The booklet plan shall consist of three portions. The first portion, front matter, shall consist of general information. The second portion, terminal table, shall consist of all sheets containing the terminal data of the equipment or systems units, junction and connection boxes, and switches (isolation and selection). The third portion, wire table, shall consist of all sheets containing interconnection wiring arranged by cable number, followed by an interconnection table of all jumpers, and the system aspect consisting of a schematic drawing of the sonar-fire control interface, power, ships gyro, pitometer log, and all other external connections.

3.3 Arrangement of sheets.- The arrangements of the booklet plan sheets shall conform to the requirements specified hereinafter.

3.4 Front matter.- The front matter of the booklet plan shall consist of the following sheet items:

(a) Title sheet.
(b) Table of contents.
(c) Alteration table and reference drawings.
(d) General notes.
(e) Wire dictionary.
(f) List of material and unit number designations.
(g) Abbreviations.
(h) Cable table.
(i) Power table.

3.5 Terminal table.- The second portion of the booklet plan, terminal table, shall conform to the requirements specified herein.

3.5.1 Sheet content.- Each sheet of the required number of sheets shall contain the tabular terminal data for one unit terminal board or jack only, except where these units are small enough and have relatively few connections so that several such units may go on one sheet. For example; the connections of all sonar loudspeakers on one sheet or the connections of all sonar dial indicators on one sheet. The first sheet of each group is the lead sheet and contains the tabulated cabling information of that unit or set of units.

3.5.2 Order of sheets.- The sheets contained in the second portion of the booklet plan shall be arranged in the order of the importance of the unit to the equipment or system and the sequence of restoration of service for damage control. The order of arrangement of the equipments or systems or both and related units in the plan will be in the order of the importance to the sonar platform. Where any single unit necessarily has more than one sheet to make up the whole, the sheets should be grouped together. The arrangement is as follows:

(a) Search sonar.
    (1) Console.
    (2) Sonar signal amplifiers.
    (3) Sonar signal receivers.
    (4) Transducer.
    (5) Computing units.
    (6) Transmitter.
    (7) Remote indicators.
    (8) Loudspeakers.
(b) Resolving equipment.
    (1) Console.
    (2) Resolving units.
    (3) Other associated units.
(c) Depth determining or attack sonar. Order of sheets to be same as listed for search sonar.
(d) Communications sonar.
    (1) Control.
    (2) Transmitter-receiver.
    (3) Transducer.

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(e) Sonar IFF and classification.
(f) Tactical aids.
(g) Junction or connection boxes.
(h) Switches, isolation (ACO) and selection.
(i) Incidental units (mark/track switches, mark indicators and others).

3.6 Wire table.- A part of the third portion of the booklet plan, wire table, shall conform to the requirements specified herein.

3.6.1 Sheet content.- Each sheet shall contain the tabular wire data of as many numerically arranged cables as required to fill the sheet. Where any single cable necessarily has more than one sheet to make up the whole, the sheets should be grouped together.

3.6.2 System aspects.- The other part of the third portion of the booklet plan, system aspects, shall conform to the requirements specified herein.

3.6.2.1 Sheet content.- Each sheet shall consist of a schematic drawing of the sonar interface as presented to the fire control system. Only those circuits required to portray clearly the sonar/fire control information transfer shall be shown. Synchro bus designators shall be shown for a standard CCW order. For counterclockwise rotation of the synchro when viewed from the shaft end, the stator windings S1, S2, S3 should be connected respectively to 1MB1, 1MB2, 1MB3 or the phase rotation of the ships wiring. For clockwise rotation, the S1 and S3 leads are interchanged. Sheets consisting of schematic drawings for all of the system aspects or external connections to the sonar gear or system, shall follow those for the sonar/fire control interface.

3.7 Plan title.- The plan title shall conform to the layout as shown on figure 1. Lettering shall be in accordance with 3.12.1.

3.8 Security Classification.- Every effort shall be made to exclude classified information from the booklet plan. If classified information is required, the booklet plan shall be classified in accordance with the information contained therein, and safeguarded as set forth in DD-441 (Security Agreement) and its attachment, Industrial Security Manual for Safeguarding Classified Information. The security classification shall be shown on each sheet of the booklet plan. The degree of classification for the booklet will be determined by the highest degree of security classification placed on any one portion of the material contained in the booklet.

3.9 Sheet contents.- The booklet plan shall contain all items listed herein and shown on the format sheets; and shall conform to the applicable requirements specified hereinafter.

3.9.1 Submission of the booklet plan wire tables; modifications and revisions.- In general the booklet plan wire tables for the basic sonar gear or system, its modifications and revisions, will be submitted in accordance with MIL-D-23140. For the purpose of distinguishing between modifications and revisions as applicable to the wire tables for the basic sonar system, definitions are as follows:

Modifications - Those changes to a sonar system of a major nature which add to or modify the system to provide or delete one or more important operational functions, and usually involving whole units and many external interconnecting cables.

Revisions - Those changes to a sonar system in the nature of corrections or minor field changes which are considered an enhancement of the operational functions already provided, and which usually do not involve the addition or deletion of any units or external cables.

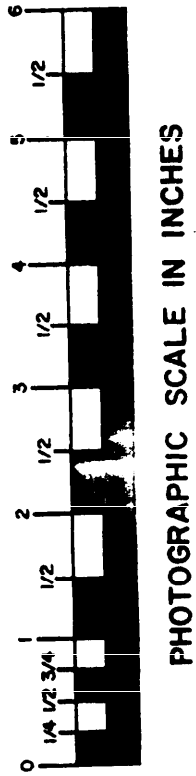
Revisions or changes shall be submitted as they become available. Modifications shall be submitted as a package or complete set and the sheets numbered as specified in 3.10.3 and letter suffixed as specified in 3.10.4. Note that the letter suffixes are shown only for examples and are not to be construed as the ones to be finally used by the contractor in making up the wire tables. If the contractor is in doubt or wishes to confer with the Bureau of Ships on letter usage, he should contact Code 679C4.

3.10 Revisions.- All revisions and corrections to booklet plan sheets shall be entered on the ALTERATION TABLE in brief form. All notations entered into the ALTERATION TABLE of revisions and corrections to the booklet plan shall be accompanied by the alteration designation of the booklet plan, the authorization, the date the revisions and corrections were made, and the initial of the person responsible. All alterations to

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TITLE SHEET LAYOUT

220	C	D								
210	A	A	B	C	A	A	B	B	B	B
200	B	B	B	B	B	B	E	E	E	E
190	E	B	B	B	F	F	F	A	A	A
180	A	G	G	G	E	F	B	B	B	B
170	C	C	C	D	A	A	A	A	A	A
160	B	C	D	D	A	A	D	C	D	C
150	A	A	A	A	A	B	B	A	A	A
140	A	A	A	A	A	A	B	B	B	C
130	C	C	C	D	D	A	B	C	C	C
120	F	F	G	E	E	E	D	A	A	A
110	A	B	C	A	A	A	A	A	A	A
100	B	B	D	A	A	A	A	A	A	C
90	C	C	C	C	C	C	C	D	C	C
80	C	D	E	E	G	G	G	G	A	A
70	C	C	C	C	D	E	E	G	A	B
60	A	B	B	B	B	B	A	A	A	A
50	A	A	A	A	A	A	A	A	A	A
40	C	C	C	A	B	C	D	E	C	C
30	D	B	B	B	C	A	A	C	B	A
20	C	A	A	C	A	B	B	A	A	A
10	A	A	A	A	A	B	B	B	C	E
0		H	A	B	A	A	B	C	A	B
	0	1	2	3	4	5	6	7	8	9



BUREAU OF SHIPS SIGNATURE	
OFFICER IN CHARGE	
HEAD ENGINEER	
PROJECT ENGINEER	
CHECKER	
DRAFTSMAN	
DATE	
FOR CHIEF OF BUREAU	
DATE	
FEDERAL SUPPLY CODE	
CONTR. DRAWING NO.	

NAVY DEPARTMENT — WASHINGTON, D.C.

BUREAU OF SHIPS

NOMENCLATURE
TYPE OR MODEL NUMBER
TYPE OF PLAN

CODE IDENT. NO.	SIZE	BUREAU OF SHIPS DRAWING NO.	REV.
	A	RE	
SCALE:	SHEET:		

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booklet plan shall be designated by an upper case letter, and each alteration shall cause the next letter in the alphabet to be entered into the booklet plan in the required locations listed hereinafter. The revision designation is to be done in pencil on the final linen.

TITLE SHEET, title block: The latest revision designation for any sheet in the plan.
TITLE SHEET, revision table: The latest revision designation for the particular sheet.

ALTERATION TABLE: The latest revision designation for any sheet in the plan.
Each affected sheet, title block.

3.10.1 Deletions.- Change sheets received with later or proper revision letters and with numbers exactly matching the numbers of the installation plan booklet sheets now in the field, will replace the sheets in the field. No other sheets shall be discarded in entering changes, unless instructed to do so. When sheets of the installation plan booklet are deleted and an insufficient number of replacements are inserted such that a gap exists in the continuity of sheet numbering, a plain sheet shall be inserted bearing the next page number and a note indicating the sheet or sheets deleted and the identity of the next effective sheet.

3.10.2 Additions.- When an added sheet is a replacement page, it will bear the same number as the replaced sheet with the proper revision letter. Sheets with page numbers having alphabetical suffixes shall be used with special installation configurations or modifications. These can be added to the masters for the installation plan booklet in their proper place, or kept in a separate packet or folder for use as the situation demands. The latter will usually be found to be the easiest in handling. When white copies are made, the replacement sheets will be filed in their proper place in the installation plan booklet.

3.10.3 Sheet numbering.- Sheets in the installation plan booklet will be consecutively numbered for the BASIC sonar system configuration. Added correction sheets to the basic booklet will be indicated by a dash (-) and a number added, thus:

127-1, 127-2, 404-2, etc.

If the sheet in the booklet is to be directly replaced, the replacement sheet will bear the same number, but will usually have a later sheet revision letter. Sheets added to the correction sheets will be numbered using the point system, thus:

127-1.1, 127-1.2, 127-2.1, 404-1.1, etc.

3.10.4 Letter suffixes.- Letter suffixes to the basic booklet sheet numbers will be used to denote a different configuration or major modification of the same basic sonar system. Sheets numbered in this manner will be used instead of a similarly numbered sheet lacking the letter suffix, for the sonar system configuration to which the letter applies. The basic sheet will not be discarded unless the ship getting this configuration desires to do so. The use of letter suffixes in sheet numbering is shown as follows and on figure 2:

Sheet No.		Meaning of Use
127	(Rev. A)	A sheet number of a basic booklet.
127	(Rev. B, C, etc.)	Subsequent revisions and replacements to sheet no. 127 Rev. A.
127-1	(Any Rev.)	Additional sheet to basic booklet.
127-1.1	(Any Rev.)	Additional sheet to basic booklet between sheet 127-1 and 127-2 or next sheet.
127A	(Any Rev.)	Sheet to be used in place of sheet 127 when ship has ASPECT modification.
127A-1	(Any Rev.)	Sheet added to the ASPECT change. Can also replace sheet 127-1 for use with ASPECT (see instruction sheet).
127AB-1	(Any Rev.)	Sheet added to the ASPECT change with Dynamic Braking. Can also replace 127-1 for use in this configuration. (See instruction sheet).

NOTE: No sheet is to be discarded unless it bears the identical page number to a sheet included with the modification change sheets.

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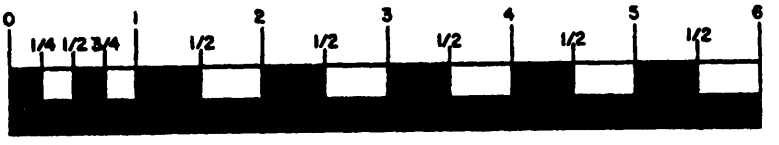
MODIFICATION CHANGE INSTRUCTIONS

SUFFIX LTR.	TYPE OF INSTALLATION
None	Basic AN/SQS- sonar system
A	ASPECT only added
B	Dynamic Braking only added
C	RDT only added
AB	with ASPECT and Dynamic Braking
BC	with Dynamic Braking and RDT
AC	with ASPECT and RDT

NOTES:

1. Use the sheets with the indicated letter suffix to the page numbers for the designated installation.

2. For the modification indicated, the letter suffix page numbered sheets exactly replace those sheets in the basic tables which have identical numerical page numbers. The dashed (-) page numbers are additional sheets if not replacing a basic sheet with the same dashed page number.



PHOTOGRAPHIC SCALE IN INCHES

CODE IDENT. NO.	SIZE	BUREAU OF SHIPS DRAWING NO.	REV.
	A	RE	
SCALE:	SHEET:		

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Figure 2 - Modification change instruction sheet example.

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3.10.5 Format.- The general format and content of the instruction sheet to be added to the installation plan booklet is shown on figure 2. The example used is not to be confused with any existing sonar system and is shown just to indicate where the letters could denote the system configuration, but it is cautioned that the selection of letters is left to the contractor's discretion, subject to approval by EMEC (Electronic Maintenance Engineering Center). The instruction sheet or sheets will also indicate as notes any special cases such as a mod. A installed prior to a mod. B, or a mod. B installed prior to a mod. A, which would entail a different set of change sheets for each condition.

3.10.6 Alterations.- The method of indicating the alteration revision letters for sheets with original numbers is shown in the example, figure 1. The methods used for indicating the alterations or revision letters of added sheets for the above system of numbering, are shown in the example, figure 3. Blank spaces in the alteration tables on the title pages indicate that the sheet (or sheets) for that page number was deleted.

3.11 Layout.- The layout of all sheets comprising a booklet plan shall conform to the layout sheets specified herein, and shall have the general appearance of the figures, tables and examples specified hereinafter. The sheets shall conform to the requirements specified herein.

3.12 Title sheet.- The title sheet shall be constructed so as to agree with the layout sheet and the example, and shall contain all the properly distributed information shown on figure 1.

3.12.1 Lettering.- The lettering sizes shall be as tabulated:

TITLE	#200 LeRoy template or equal
SECURITY CLASSIFICATION	#240 LeRoy template or equal
SECURITY INFORMATION	#120 LeRoy template or equal
Revisions	#100 LeRoy template or equal

All other lettering will be done with #120 LeRoy template or equal, except where space will not permit.

3.13 Table of contents.- The table of contents sheet shall be constructed as shown on table I.

3.13.1 Contents.- All sheet titles in the contents list shall be in ascending order of the sheet numbers assigned. The first portion of the list shall be sub-titled front matter, and shall contain the sheet titles as set forth under arrangement of sheets. The second portion of the list shall be sub-titled terminal table, and shall consist of the titles of all tabular terminal data sheets as set forth in arrangement of sheets.

3.13.2 Lettering.- In the first issue of the wire tables for a sonar system, the lettered content, if typewritten, shall be 12 Point Bold, consistent throughout in either Executive, Sans Serif (Gothic), or lettered using size tabulated below:

Table of contents	#120 LeRoy template
All other lettering	#100 LeRoy template

3.14 Alteration table.- The alteration table shall be constructed as shown on figure 4.

3.14.1 Contents.- Revisions to the plan shall be entered into the sheet in accordance with revisions (see 3.10). All references to drawings will be listed under reference drawings, same table.

3.14.2 Lettering.- Lettering shall be as specified in 3.13.2.

3.15 Notes.- The notes sheet shall be constructed as specified in table II.

3.15.1 Contents.- This sheet shall contain all general notes necessary to clarity or to add pertinent information to the booklet plan.

3.15.2 Lettering.- Lettering shall be as specified in 3.13.2.

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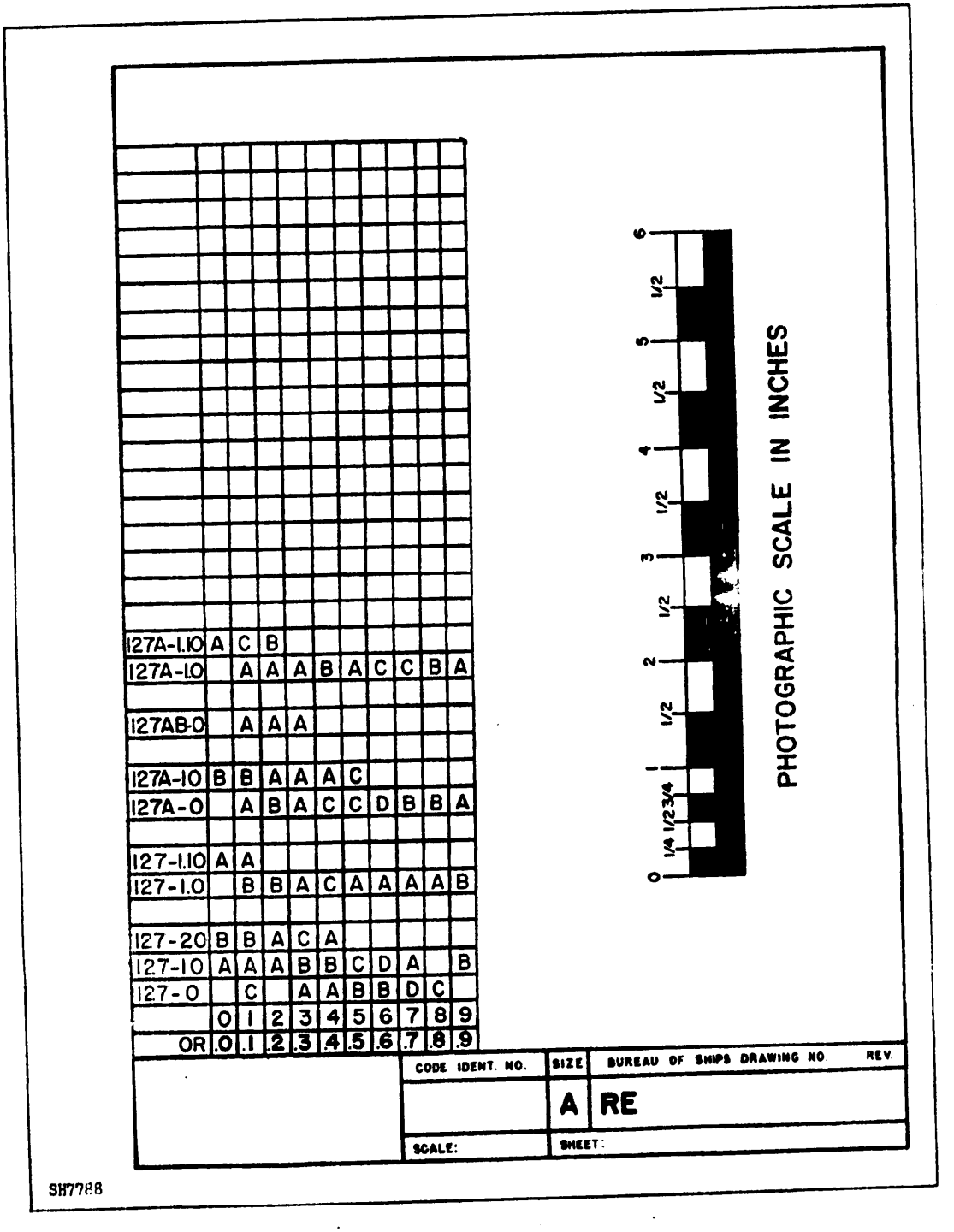


Figure 3 - Method of indicating alterations, examples.

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Table I - Table of contents.

TABLE OF CONTENTS

FRONT MATTER

TITLE SHEET(S)	1, 2
TABLE OF CONTENTS	3, 4
ALTERATIONS AND REFERENCE TABLE	5
GENERAL NOTES	6
WIRE DICTIONARY	7, 8
LIST OF MATERIAL AND UNIT NUMBER DESIGNATIONS	9, 10
ABBREVIATIONS	11, 12
CABLE TABLE	13-20
POWER TABLE	21

TERMINAL TABLE

CONNECTION BOX	22-25
CIRCUIT BREAKER	26
DUMMY LOAD	27
JUNCTION BOX - A	28-39
LOUDSPEAKER	40
CONTROL INDICATOR	41-63
DATA CONVERTER	64-78
RECEIVER SCANNING SWITCH	79-88
A F AMPLIFIER	89-96
TRANSDUCER TRANSFER SWITCH	97-140
TRANSMITTER	141-154
TRANSDUCER	155-181
SONAR TEST SET	182, 183
VOLTAGE REGULATING TRANSFORMER	184
CONNECTION BOX R-5K1	185-230
R-5K2	231-276
CAN-55134B TACTICAL RANGE RECORDER	277

WIRE TABLE

CABLES	278-398
JUMPERS	399
SYSTEM ASPECTS	400-412

CODE IDENT. NO.	SIZE	BUREAU OF SHIPS DRAWING NO.	REV.
	A	RE	
SCALE:	SHEET:		

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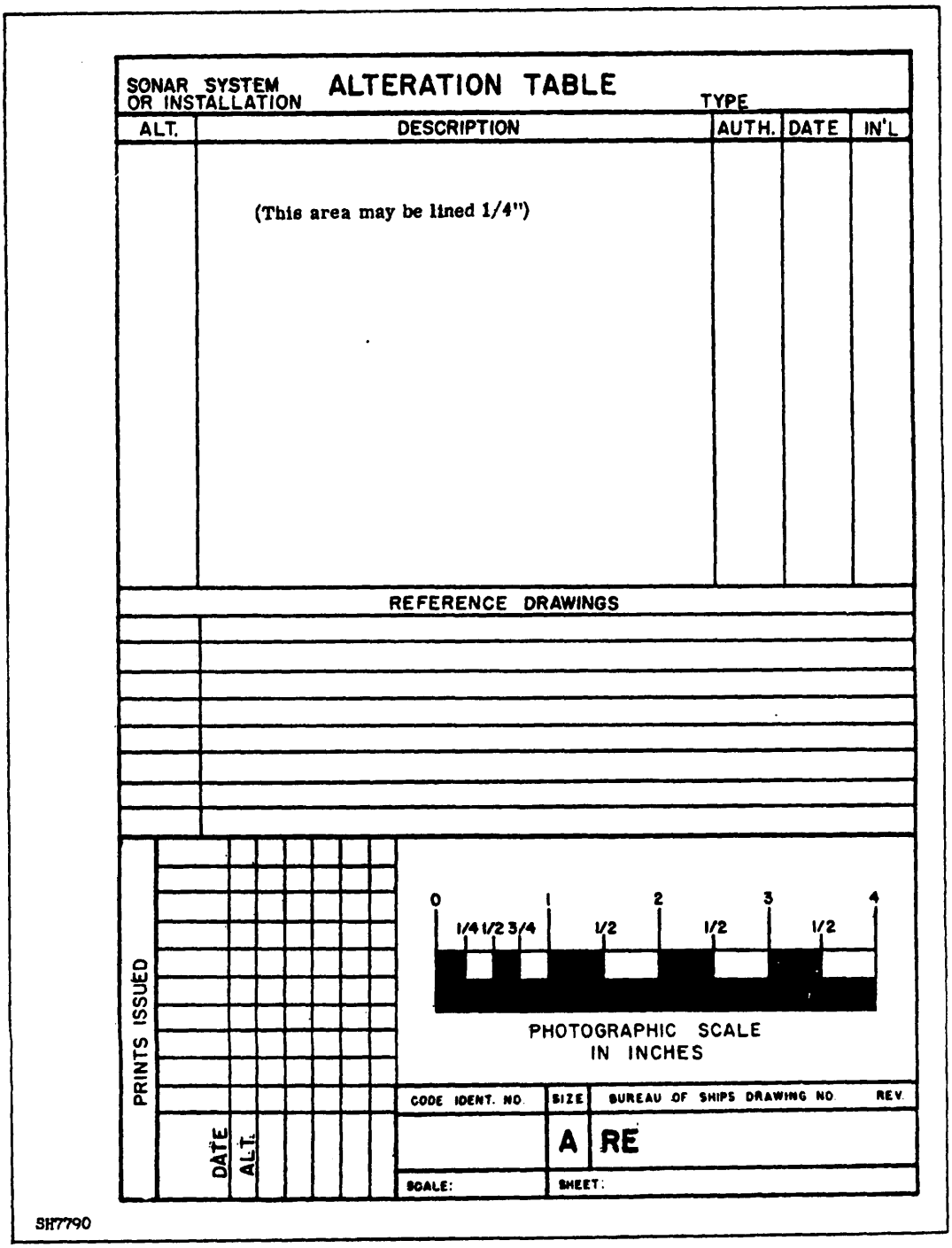


Figure 4 - Alteration table layout.

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Table II - General notes sheet.

GENERAL NOTES

1. ALL ELECTRONIC EQUIPMENTS IN THIS SYSTEM ARE TO BE GROUNDED PER BUSHIPS INSTRUCTION 9670.16.

2. SHIELDS ARE TO BE SECURELY CONNECTED TO THE TERMINALS INDICATED. A DASH (-) ADJACENT TO THE "SHLD" DESIGNATION IN BOTH THE TERMINAL AND WIRE TABLES INDICATES THE SHIELD IS DEAD ENDED AS NEAR AS POSSIBLE TO THE TERMINAL BOARD. CARE SHOULD BE TAKEN TO INSURE ADEQUATE INSULATION FROM GROUND AND OTHER CONDUCTORS.

3. THE TRANSDUCER TRANSFER SWITCHES ARE DELIVERED WITH INTERNAL JUMPERS AND RESISTORS INSTALLED. ALL OTHER JUMPERS IN THE SYSTEM ARE TO BE FURNISHED BY THE INSTALLING ACTIVITY.

4. AN ASTERISK SUCCEEDING A WIRE NUMBER IN THE TERMINAL TABLE INDICATES INFORMATION FROM THAT WIRE IS ENTERING THE TERMINAL.

5. ALL WIRES AND CABLES ARE TO BE MARKED ACCORDING TO THIS SYSTEM PLAN. WIRE DESIGNATIONS ARE TO BE STAMPED ON SYNTHETIC RESIN TUBING (MIL. SPEC. MIL-I-631C, TYPE F, GRADE A). EACH WIRE IS TO BE EASILY ASSOCIATED WITH THE CABLE OF WHICH IT IS A PART.

6. IF REDUCED DIAMETER CABLES ARE NOT AVAILABLE, USE MHFA, DHFA, ETC., FOR MSCA, DSGA, ETC.

7. SYNCHRO RANGE TRANSMISSION TO UBFC IS ELECTRICALLY ZEROED AT 10,000 YARDS.

8. INSTALL "QG" JUMPER IN TRR; SEE INSTRUCTION BOOK.

9. MODIFY LOUDSPEAKER, TYPE 49546, IN ACCORDANCE WITH REF. 2.

10. INSTALL 155-VOLT, 6-WATT BULBS IN ALL EXTERNAL DIAL INDICATORS IN LIEU OF 115-VOLT BULBS.

11. INSTALL MATCHED (WITHIN 1/2 OF 1% OF EACH OTHER), LOW TEMPERATURE COEFFICIENT, WIRE WOUND, 680 OHM RESISTORS WYE CONNECTED TO TERMINALS 1A-17, 1A-18, AND 1A-19.

12. SYNCHRO DEPTH TRANSMISSION TO UBFC IS ELECTRICALLY ZEROED AT 50 FEET.

13. SEE INSTRUCTION BOOK MANUAL FOR LOADING TERMINALS 15A-4 THROUGH 15A-6 AND REZEROING THE AN/SQR-8 SWEEP GENERATOR.

CODE IDENT. NO.	SIZE	BUREAU OF SHIPS DRAWING NO.	REV.
	A	RE	
SCALE:	SHEET:		

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3.16 Wire dictionary.- The wire dictionary sheet shall be constructed as specified in table III.

3.16.1 Contents.- The finished wire dictionary shall be derived from the key dictionary for wire numbering and shall include only those designations pertaining to this equipment or system booklet plan.

3.16.2 Lettering.- Lettering shall be as specified in 3.13.2

3.17 List of material and unit number designations.- The list of material and unit number designations shall be constructed as specified in table IV.

3.17.1 Contents.- The sheet shall contain the unit number designation for the specific sonar equipment, quantity, name, type designation, and furnishing activity.

3.17.2 Lettering.- Lettering shall be as specified in 3.13.2.

3.18 Cable table and abbreviations.- The cable table and abbreviations sheets shall conform to the general appearance as specified in tables V and VI.

3.18.1 Contents.- The cable table shall contain all cables used in interconnecting systems, equipments, units, and other items in the plan. The cables shall be grouped by identity, and shall be in order of ascending numbers. Units connected, cable types and sizes, and the number of active wires in each cable shall be shown in the table. The abbreviation sheet shall contain all abbreviations with their meaning used in the plan, and shall be in alphabetical order. Where pairs or triads are indicated in the ACTIVE column of the cable table, the number shall be followed by "pr" for pairs and by "tr" triads. If the contractor elects to use the terminal boards to which the cables are fanned out when indicating cable runs between units in the cable table, he shall be consistent and use this method throughout the table.

3.18.2 Lettering.- Lettering shall be as specified in 3.13.2.

3.19 Power table.- The power table sheet shall be constructed as specified in table VII.

3.19.1 Contents.- The power table shall contain the name of all units connected to primary power bus, all primary power carriers (cable number), the power load on each power carrier, the voltage on each power carrier, and the fuse size.

3.19.2 Lettering.- Lettering shall be as specified in 3.13.2.

3.19.3 Other tables.- Other tables considered of sufficient importance to be included shall follow this table. Some of these are tables of special connectors plugs and cables, table of equivalent nomenclatures, special color coding (transducer cables and synchro leads), and a table of special precautions or notes applicable to the installation.

3.20 Terminal table lead sheets.- Each lead sheet shall be constructed as specified in table VIII.

3.20.1 Contents.- The lead sheet shall contain the unit name, cable numbers, cable types, and opposing units.

3.20.2 Lettering.- Lettering shall be as specified in 3.13.2.

3.21 Terminal table.- Each terminal table sheet shall be constructed as specified in table IX.

3.21.1 Contents.- Each terminal table sheet shall contain the unit name, function of signal existing at each terminal, terminal numbers, wire numbers (developed from wire dictionary), cable numbers, and opposing units and terminals. Asterisks shall be used to indicate input signal leads. The addition of WIRE COLOR and GROUP NUMBER columns to the cable table is permissible but not mandatory except to determine specific wires in special cables (transducer) and to separate wire groups which are critical with respect to interference, if no other method of indication is used.

3.21.2 Lettering.- Lettering shall be as specified in 3.13.2.

3.22 Terminal cable drawings.- Each terminal table drawing shall be constructed as shown on figure 5.

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Table III - Wire dictionary.

WIRE DICTIONARY
WIRE NUMBERS ARE CODED AS FOLLOWS:

SAMPLES:	(R)	1	SK		1
	(R)	24	SK	3	024
REF. PART:	a	b	c	d	e

(IN THE FIRST SAMPLE, ONE LEG OF AC POWER IS INDICATED DIRECTLY CONNECTED. THE MISSING ZEROS FOR d AND e ARE OMITTED. IN THE SECOND SAMPLE, A Ba SYNCHRO ORDER IS INDICATED AFTER PASSING THROUGH THREE MODIFICATIONS. THE FIRST ZERO FOR d IS OMITTED).

PART a: THE "R" DESIGNATES ELECTRONICS AND IS OMITTED FROM THE WIRE NUMBER IN THIS BOOKLET.

PART b: DIGITS FROM 1 TO 99 INDICATE THE PRIMARY FUNCTION OF THE WIRE AS FOLLOWS:

1.	PRIMARY SHIP'S AC POWER
2.	SECONDARY AC POWER (MODIFIED BY TRANSFORMERS SWITCHED, REGULATED, ETC.)
3.	DC SUPPLIES
4.	CIRCUIT ENERGY SIGNALS FOR STIMULATION, REGULATION, OR EXCITATION OF ELECTRONIC CIRCUITS (TRIGGER PULSES, SWEEP SIGNALS, TVG, RCG, ETC.)
5.	R. F. SIGNALS (TRANSDUCER SIGNALS).
6.	OUTPUT SIGNALS (AUDIO, VIDEO).
7.	R. F. SIGNALS, VARIABLE PLANE.
8-11.	UNASSIGNED.
12.	AC CONTROL TO RELAYS, INDICATOR, LAMPS, MOTOR STARTERS, REMOTE OR LOCAL OPERATION OF DISASSOCIATED CIRCUITS.
13.	DC CONTROL TO RELAYS, INDICATOR LAMPS, MOTOR STARTERS, REMOTE OR LOCAL OPERATION OF DISASSOCIATED CIRCUITS.
14-20.	UNASSIGNED.
21.	SYNCHRO ORDER Bya.
23.	SYNCHRO ORDER qBya, iBya, icBya.
24.	SYNCHRO ORDER Ba.
26.	SYNCHRO ORDER Bda'.
31.	SYNCHRO ORDER Ra.
33.	SYNCHRO ORDER qRa.
34.	SYNCHRO ORDER Rha.
41.	SYNCHRO ORDER Eua.
42.	SYNCHRO ORDER Edua'.
71.	SYNCHRO ORDER Co.
72.	SYNCHRO ORDER DMho.
78.	ZERO ORDER.

CODE IDENT. NO.	SIZE	BUREAU OF SHIPS DRAWING NO.	REV.
	A	RE	
SCALE:	SHEET:		

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Table III - Wire dictionary (Cont'd).

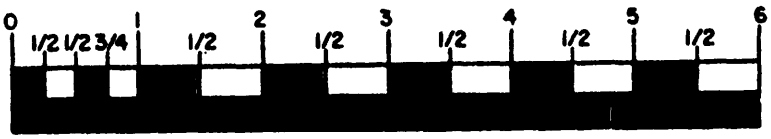
WIRE DICTIONARY (CONT'D)

PART c: THE "S" SERIES OF NUMBERS INDICATE SPECIFIC SONAR EQUIPMENT AS FOLLOWS:
SD - DEPTH DETERMINING
SK - AZIMUTH SCANNING
SP - AUXILIARY, ATTACK AIDS

PART d: DIGITS FROM 00 TO 99 INDICATE THAT THE WIRE FUNCTION HAS BEEN MODIFIED BY AN "OPEN CIRCUIT/CLOSED CIRCUIT" CIRCUIT ELEMENT (SWITCHES, FUSES, RELAY CONTACTS).

PART e: DIGITS FROM 000 TO 999 TO INDICATE SPECIFIC WIRE FUNCTIONS.

NOTES:
1. RETURN CIRCUITS, SUCH AS FROM THE TRANSDUCER, USE THE DESIGNATION "SKK" TO ASSOCIATE PAIRED WIRES AND REDUCE THE NUMBERS USED.
2. A FINAL DIGIT OF "0" ORDINARILY INDICATES A GROUND CONNECTION. "4SK10" HAS BEEN USED TO INDICATE THE SYSTEM GROUND.
3. ONLY THE WIRE FUNCTIONS APPEARING IN THIS PARTICULAR INSTALLATION PLAN BOOKLET ARE LISTED ABOVE IN PART b



PHOTOGRAPHIC SCALE IN INCHES

NOTE: THE PHOTOGRAPHIC SCALE IS REPEATED ON ALL FIGURES AND TABLES WHERE SPACE IS AVAILABLE TO INDICATE THE AMOUNT OF PHOTOGRAPHIC REDUCTION FROM ACTUAL SIZE. THE SCALE IS INCLUDED ONLY FOR COMPARISON AND IS NOT TO BE SHOWN ON FINISHED SHEETS.

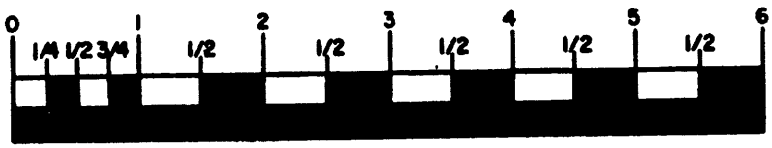
CODE IDENT. NO.	SIZE	BUREAU OF SHIPS DRAWING NO.	REV.
	A	RE	
SCALE:	SHEET:		

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LIST OF MATERIAL AND UNIT NUMBER DESIGNATIONS

UNIT NO.	UNIT	GF*	FIA**
	CONNECTION BOX		1
	CIRCUIT BREAKER		1
9	DUMMY LOAD, DA-167/SQS	1	
	JUNCTION BOX - A		1
1	CONTROL INDICATOR, C-1022/SQS	1	
2	DATA CONVERTER, CV-108/SQS	1	
3	RECEIVER SCANNING SWITCH, SA-161/SQS	1	
4	A F AMPLIFIER, AM-104/SQS	1	
5	TRANSDUCER TRANSFER SWITCH, SA-200/SQS	1	
8	TRANSMITTER, T-601/SQS	1	
11	TRANSDUCER, TR-109/SQS	1	
13	SONAR TEST SET, TS-1777/SQS	1	
14	VOLTAGE REGULATING TRANSFORMER, TF-608/SQS	1	
15	CONNECTION BOX R-SK1	1	
16	CONNECTION BOX R-SK2	1	
	TACTICAL RANGE RECORDER, CAN-55134B		1
10	LOUDSPEAKER, LS-218/SQS	1	



PHOTOGRAPHIC SCALE IN INCHES

NOTE: THE PHOTOGRAPHIC SCALE IS REPEATED ON ALL FIGURES AND TABLES WHERE SPACE IS AVAILABLE TO INDICATE THE AMOUNT OF PHOTOGRAPHIC REDUCTION FROM ACTUAL SIZE. THE SCALE IS INCLUDED ONLY FOR COMPARISON AND IS NOT TO BE SHOWN ON FINISHED SHEETS.

CODE IDENT. NO.	SIZE	BUREAU OF SHIPS DRAWING NO.	REV.
	A	RE	
SCALE:	SHEET:		

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Table V - Cable table.

CABLE TABLE

NUMBER	FROM	TO	TYPE	ACTIVE	
R-SB5	UQC CONT	DD PANEL	TTHFWA-1-1/2	1	PR
R-SD1	UQN CONT	TRANSDUCER	DSGA-4	2	
R-SK01	VR	PWR PANEL	DSGA-23	2	
1	CB R-SK-3	SQS CONT INDICATOR	DSGA-9	2	
2	CB R-SK-3	SQS DC	DSGA-9	2	
3	SQS CONT INDICATOR	VR XFORMER	FSGA-4	4	
4	CB R-SK-3	SQS SSA	DSGA-9	2	
5	CB R-SK-3	AF AMP	DSGA-4	2	
6	SQS CONT INDICATOR	TTS CONT	MSCA-19	15	
R-SL1	CB R-SK-3	SQR CONT INDICATOR	DSGA-9	2	
2	CB R-SK-3	SQR DATA COMPUTER	DSGA-4	2	
3	CB R-SK-3	SQR RCVR SSA	DSGA-9	2	
4	XDUCER	2X SW	TSP-31	31	PR
5	XDUCER	3X SW	TSP-31	28-1/2	PR

CODE IDENT. NO.	SIZE	BUREAU OF SHIPS DRAWING NO.	REV.
	A	RE	
SCALE:	SHEET:		

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Table VI - Abbreviations used in cable, terminal and wire tables.

ABBREVIATIONS USED IN CABLE, TERMINAL AND WIRE TABLES

ABBREVIATION	STANDS FOR
A. F.	AUDIO FREQUENCY
AMP	AMPLIFIER
ASSY	ASSEMBLY
AZ	AZIMUTH
BR	BEARING
BRK	BREAKER
CB	CIRCUIT BREAKER
CKT	CIRCUIT
COMM	COMMON
COMM., IND	COMMUTATOR, INDUCTIVE
CONT	CONTROL
CONVTR	CONVERTER
DIST	DISTRIBUTION
DOP CLASS	DOPPLER CLASSIFIER
DEPR	DEPRESSION
F. C.	FIRE CONTROL
F. C. - BSS	FIRE CONTROL - BEARING SELECTOR SWITCH
GEN	GENERATOR
GND	GROUND
GRAP	GRAPHIC
IND	INDICATOR
INTERCON	INTERCONNECTION
INTLK	INTERLOCK
JUNC	JUNCTION
LT	LEFT
LGT	LIGHT
MG	MOTOR - GENERATOR
MOD	MODULATOR
ODN	OWN DOPPLER NULLIFIER
OL	OVERLOAD
OT	OVERTEMP
POT	POTENTIOMETER
PPI	PLAN POSITION INDICATOR
PWR	POWER
(R)	SIGNAL RETURN
REC	RECEIVER
RECDR	RECORDER
RECT	RECTIFIER
REF	REFERENCE
REG	REGULATOR OR REGULATED
REJ	REJECT
REM	REMOTE
RG	RANGE
RT	RIGHT

CODE IDENT. NO.	SIZE	BUREAU OF SHIPS DRAWING NO.	REV.
	A	RE	
SCALE:	SHEET:		

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Table VII - Power table.

POWER TABLE

EQPT	CABLE NO.	POWER	VOLTAGE	FUSE
VR	R-SK01	5.3 KVA	117 VAC 1Ø	60
TRR	R-SP1	0.1 KVA	117 VAC 1Ø	5
AUX PWR RLY	R-SK02	- - - - -	117 VAC 1Ø	5
UQC	R-SB01	0.15 KVA	117 VAC 1Ø	10
UQN	R-SD01	0.15 KVA	117 VAC 1Ø	15
SQR	R-SL01	0.1 KVA	117 VAC 1Ø 400 CY	5
XFORMER	R-SK03	12 KVA	440 VAC 3Ø	75

CODE IDENT. NO.	SIZE	BUREAU OF SHIPS DRAWING NO.	REV.
	A	RE	
SCALE:	SHEET:		

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Table VIII - Terminal table lead sheets.

AN/SQS- CONTROL INDICATOR (1A, 1H)

CABLE NO.	TYPE	OPP. UNIT
R-SK1	DSGA-9	CB R-SK-3
R-SK3	FSGA-4	VR XFORMER
R-SK6	MSCA-19	TSS CONT
R-SK16	MSCA-10	AF AMP
R-SK19	MSCA-14	SQS RCVR SSA
R-SK20	MSCA-30	SQS DATA CONV
R-SK21	MSCA-14	SQS DATA CONV
R-SK22	RG-79/U	SQS RCVR SSA
R-SK24	MSCA-7	XMITTER
R-SK28	MSCA-19	SQS DATA CONV
R-SK29	TTHFWA-1-1/2	SQS SPKR CONT RM
R-SK33	TTRSA-2	CB R-SK-2
R-SK35	RG-79/U	CB R-SK-2
R-SK41	MSCA-14	CB R-SK-2
R-SK42	MSCA-14	TEST SET
R-SK46	MSCA-19	DD PANEL
R-SK51	MSCA-10	Ra RPTR CIC
R-SK53	MSCA-7	DD PANEL
R-SL16	TSGA-3	SQR CONT IND
R-SP2	TTRSA-5	TRR

CODE IDENT. NO.	SIZE	BUREAU OF SHIPS DRAWING NO.	REV.
	A	RE	
SCALE:	SHEET:		

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Table IX - Terminal table sheet.

AN/SQS- CONTROL INDICATOR FUNCTION		TB1A	WIRE NO.	CABLE NO.	OPP. TERM
SONAR TRACK	R1	56*	31SK01200	R-SK46	AD SW
			31SK01200	R-SK51	RA RPTR-4
Ra REF			31SK01200	R-SK41	CB2-21
			31SK01200	R-SD16	15A-86
	R2	57*	31SKK01200	R-SK46	AD SW
			- - -	JUMP	1A-20
72000 YD CCW 1HG B-1009	S1	58	31SK00201	R-SK46	AD SW
	S2	59	31SK00202	R-SK46	AD SW
	S3	60	31SK00203	R-SK46	AD SW
2000 YD CW 1HG B-1008	S1	61	31SK00206	R-SK46	AD SW
	S2	62	31SK00205	R-SK46	AD SW
	S3	63	31SK00204	R-SK46	AD SW
AUDIO	GND	64	6SKK01001	R-SK29	SPKR
	HI	65	6SK01001	R-SK29	SPKR
		66			
GAIN CONT MUTED		67*	- - -	JUMP	1A-85
GAIN CONT	HI	68	4SK00203	R-SK46	TB6-27
	ARM	69	4SK00202	R-SK19	4A-26
	LO	70	4SK00201	R-SK19	4A-27
DEFL REF	R1	71	4SK00300	R-SK19	4A-13
	R2	72	4SKK00300	R-SK19	4A-14
VIDEO	SHIELD	73	6SKK00002	R-SK22	4A-20
	HI	74*	6SK00002	R-SK22	4A-21
SUM-DIFF CONT	R1	75	2SK03001	R-SK19	4A-30
ODN CONT	COM	76	4SKK00005	R-SK19	4A-29

CODE IDENT. NO.	SIZE	BUREAU OF SHIPS DRAWING NO. REV.
	A	RE
SCALE:	SHEET:	

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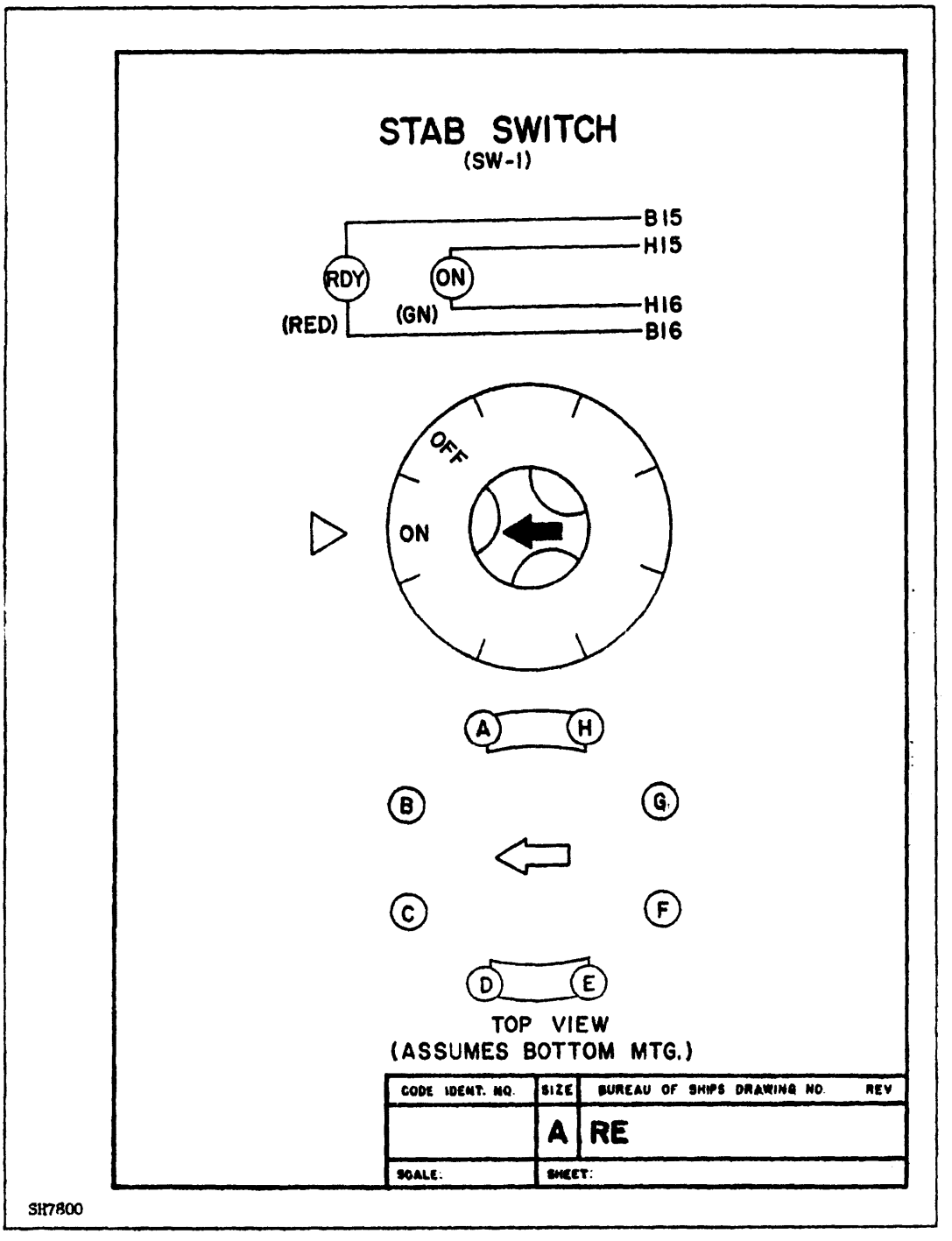


Figure 5 - Terminal table drawings.

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3.22.1 Contents.- Terminal table drawings should be kept to a minimum and only be used to portray information that cannot be fully covered by the terminal table, as transducer wiring, ACO switch orientation, and special switches and controls.

3.22.2 Lettering.- The lettering sizes shall be as tabulated:

Title . . . . . #120 LeRoy template
All other lettering . . . . . #100 LeRoy template

3.23 Wire table.- The wire table sheets shall be constructed as specified in table X.

3.23.1 Contents.- The wire table shall contain cable numbers, cable types, lead numbers, and lead terminations grouped by cable numbers that are alphanumerically arranged.

3.23.2 Lettering.- Lettering shall be as specified in 3.13.2.

3.23.3 Spare lines.- Ten percent, with a minimum of four, spare lines shall be left for additional leads following each cable group.

3.23.4 Jumper table.- The jumper table shall be constructed the same as for the wire table with the exception that there shall be no cable or wire numbers.

3.24 System aspects.- The system aspect sheet or sheets shall be constructed as shown on figures 6 and 7.

3.24.1 Contents.- The system aspects shall consist of drawings showing all information data transmitted from and received by the sonar system. Only those components directly affected by or effecting this information shall be shown.

3.24.2 Lettering.- The lettering sizes shall be as listed:

System aspects . . . . . #120 LeRoy template
All other lettering . . . . . #100 LeRoy template
All other lettering may be reduced, if necessary to . . . . # 80 LeRoy template

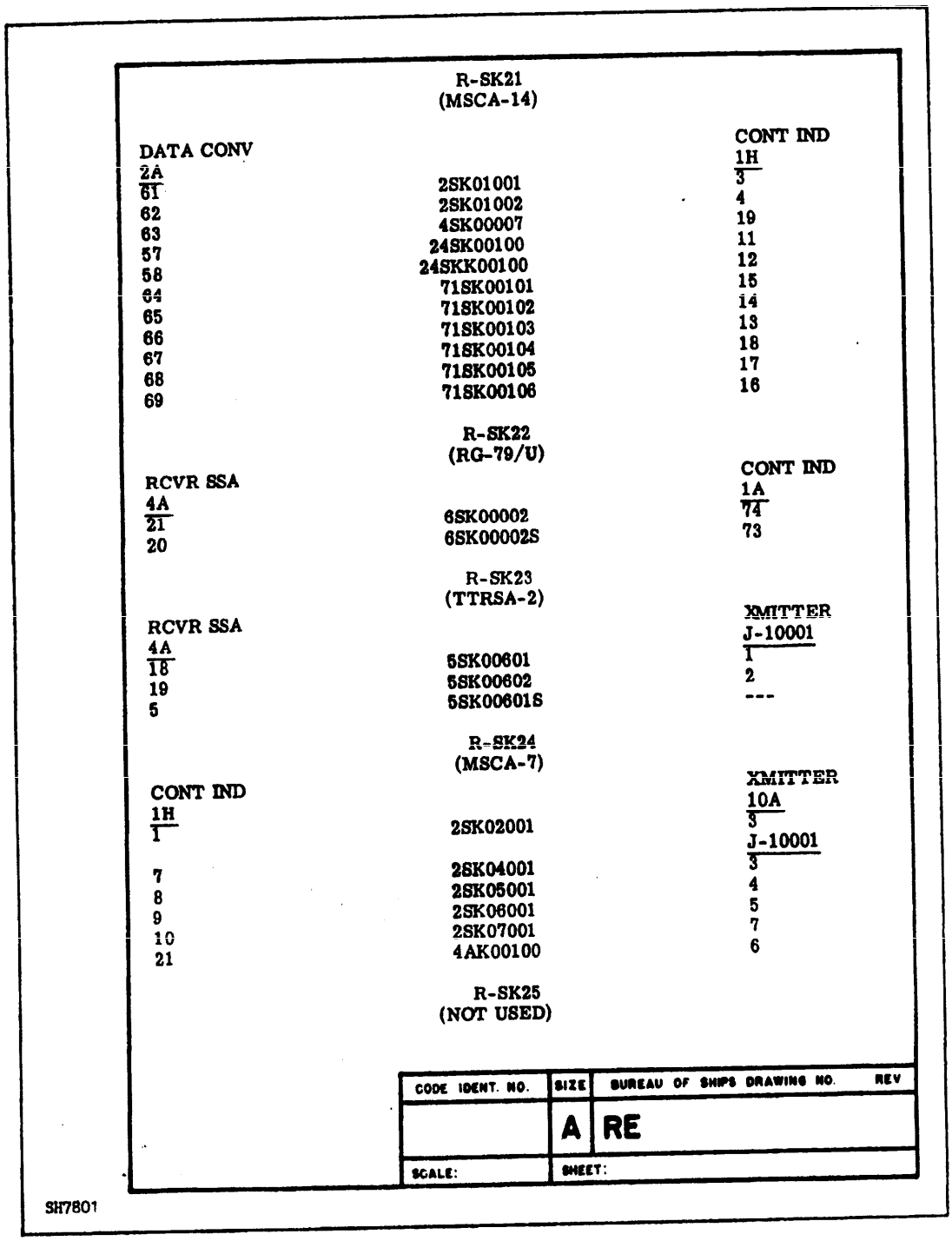
3.25 Assignment of wire numbers.- Each wire in the system shall be designated by function as derived from the key dictionary for sonar wire numbers (see 3.26), as shown on figure 8. Only wires that are always conductively coupled bear identical numbers, and these always bear identical numbers. A wire that is used for more than one function shall have a number assigned that designates the function existing when the system is in the most advanced mode of operation. Wire numbers for synchro busses shall be assigned in the proper sequence for standard CCW rotation. Wire table function columns shall indicate the function as it is used or produced within the unit, and wire numbers for functions as derived from the key dictionary shall be carried through intact if continuous from unit to unit. Thus the wire numbering sequence for standard CCW synchro rotation at one unit may indicate an apparent connection reversal if CW rotation is desired at an associated unit. Jumpers will not require wire numbers. Braids or shields for cables will be numbered as conductors using the lowest wire number of the function shielded followed by the letter "S", thus:

21SZ1S, 21SZ4S, 21SZ7S } Shields for 3 speeds
21SZ10S — Excitation shield

Where INPUT and OUTPUT are used in the function column (table IX), the word should apply to the actual use or production of that information by the unit defined in the terminal table sheet. This should not be copied verbatim from the function definition column of the associated unit on the other end of the wire. A common number for systems ground shall be assigned and used throughout the system for grounds of different functions, where feasible. For example, 4SK10 may be used to denote a system ground.

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Table X - Wire table sheets.



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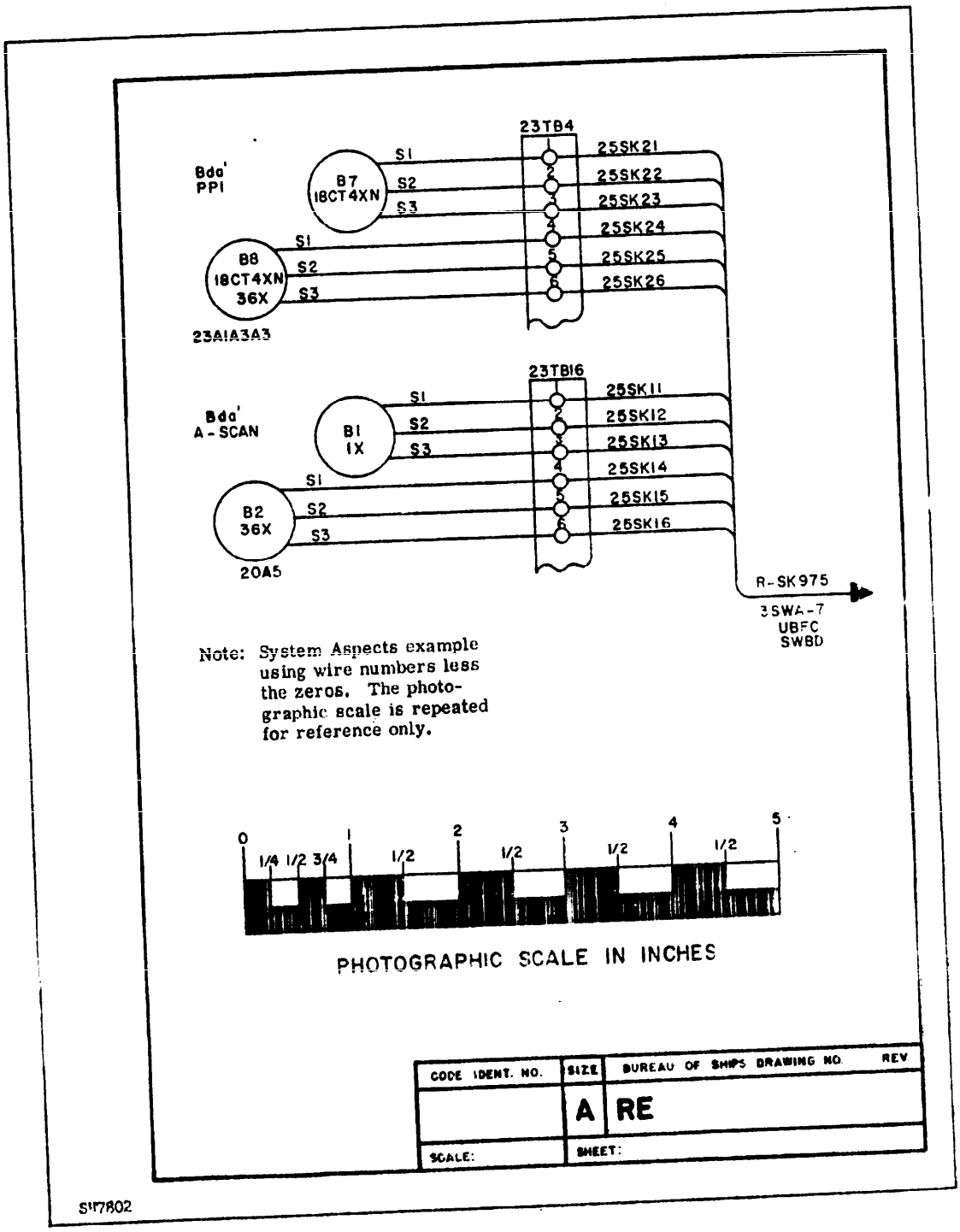


Figure 6 - System aspects.

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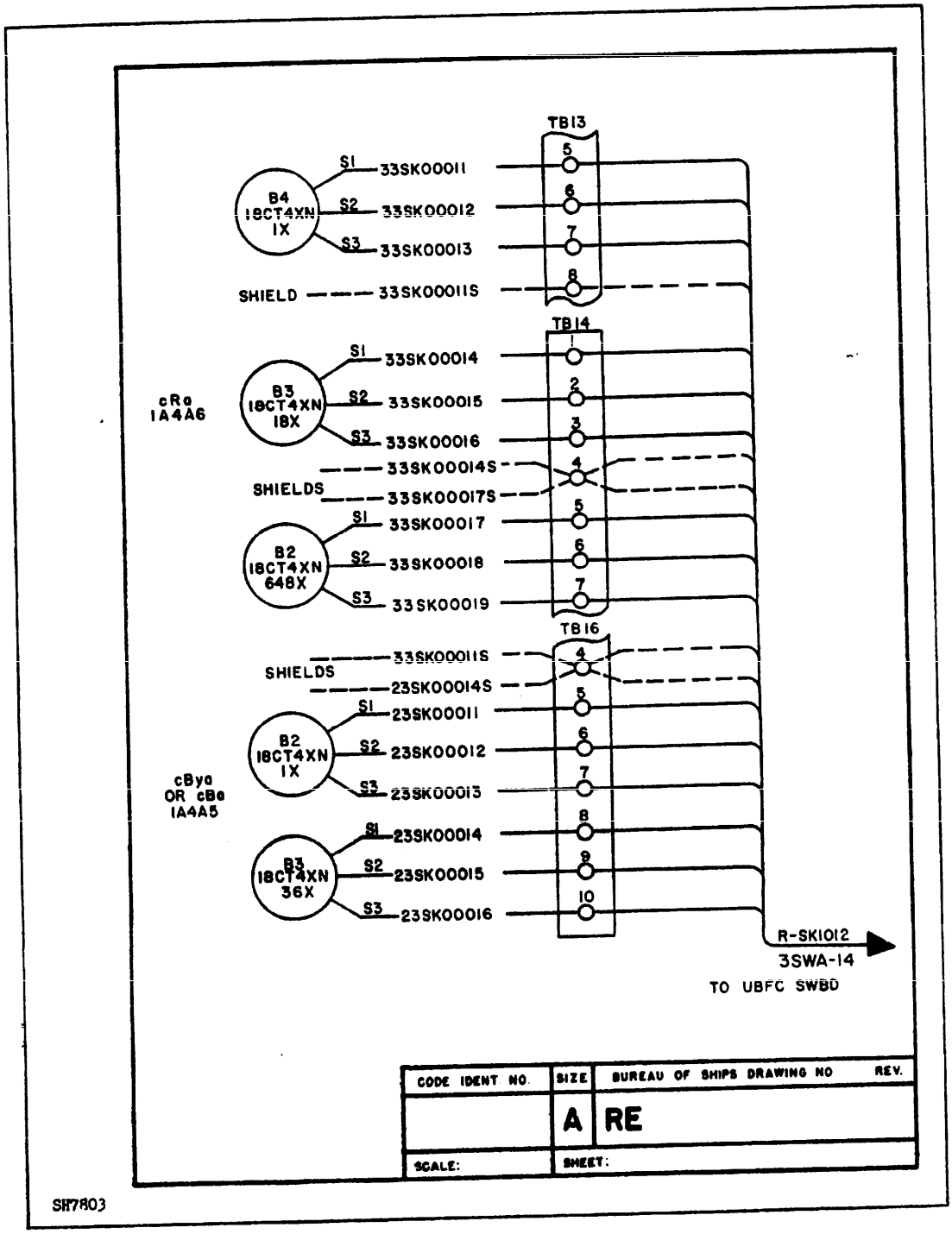


Figure 7 - System aspects (complete wire numbers).

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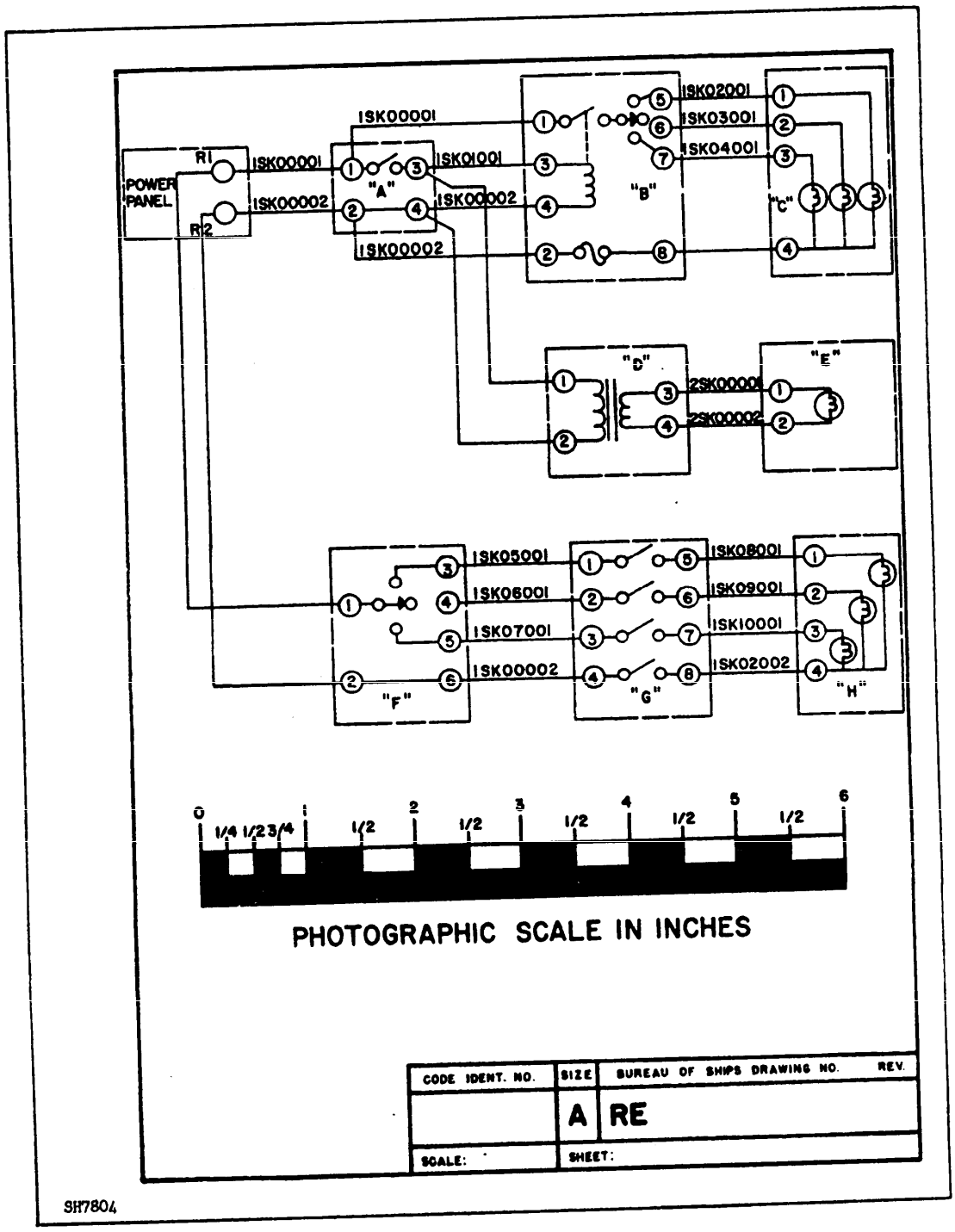


Figure 8 - Wire designations (complete wire numbers).

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3.26 Key dictionary for sonar wire numbers.- The wire dictionary will be made according to the example sheet provided. Basic wiring information will be determined from the following system:

Sample:	R	24	SK	03	004
Parts:	(a)	(b)	(c)	(d)	(e)
Possibilities:	R	1-99	SA-SX	00-99	001-999

(a) R designates electronics (the letter "R" is omitted in this booklet).
(b) Number 1 - Primary ship's AC power.
2 - Secondary AC power (primary power modified by transformers, simple circuit elements, etc.).
3 - DC Supplies.
4 - Circuit energy signals for stimulation, excitation, or regulation of electronic circuits (trigger pulses, sweep signals, TVG, RCG, blanking signals, time ticks, stepping pulses, etc.).
5 - R. F. signals, normal plane.
6 - Output signals, audio, video.
7 - R. F. signals, variable plane.
8 - Unassigned.
9 - Miscellaneous.
10-11 - Unassigned.
12 - AC control to relays, indicator lamps, motor starters, remote or local operation of disassociated circuits.
13 - DC control to relays, indicator lamps, motor starters, remote or local operation of disassociated circuits.
14 through 20 - Unassigned.
21 - Synchro order Bya.
22 - Synchro order unassigned.
23 - Synchro order q(Bya), i(Bya), ic(Bya).
24 - Synchro order Ba.
25 - Synchro order Bda.
26 - Synchro order ic(Ba).
27 - Synchro order DB.
28-30 - Synchro orders unassigned.
31 - Synchro order Ra.
32 - Synchro order unassigned.
33 - Synchro order q(Ra), DR.
34 - Synchro order Rha.
35 - Synchro order unassigned.
36 - Synchro order c(Rha).
37-40 - Synchro orders unassigned.
41 - Synchro order Eua.
42 - Synchro order Eua-E1, Edua.
43 - Synchro order Etua, i(Eua), js(Eua), Ep.
44 - Synchro order jt(Eua), jp(Eua), js(Eua), Ep.
45 - Synchro order Eua-Edua.
46-47 - Synchro orders unassigned.
48-50 - Synchro orders unassigned.
51 - Synchro order Rvua.
52 - Synchro order Rvua'.
53 - Synchro order c(Rvua).
54-55 - Synchro orders unassigned.
56 - Synchro order Bts.
57-58 - Synchro orders unassigned.
59 - Synchro order Zo.
60 - Synchro order Eio.
61 - Synchro orders unassigned.
62 - Synchro order Zd.
63-70 - Synchro orders unassigned.
71 - Synchro order Co.
72 - Synchro order DMho.
73 - Synchro order Mhoy.

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74 - Synchro order Mhox
75 - Synchro order DMho cos Ba.
76-77 - Synchro orders unassigned.
78 - Zero order.
79-99 - Synchro orders unassigned.

(c) The "S" series of letters indicate specific sonar equipments:

SA	Azimuth searchlight.
SB	Underwater communication.
SC	Computing equipment.
SD	Depth determining.
SE	Depth charge indicators.
SF	
SG	
SH	Sonar hoist/lower systems.
SI	Sonar Identification.
SJ	
SK	Azimuth scanning equipment.
SL	Listening.
SM	Monitoring system, test.
SN	Passive ranging.
SO	Bathythermograph systems.
SP	Auxiliary, attack aids.
SQ	Combination azimuth/depth.
SR	Remote indicator.
SS	Sounding, fathometer.
ST	Trainer.
SU	Object locator, mine detection.
SV	Variable depth sonar.
SW	Classification.
SX	Doppler sonar.
SZ	(Examples).

(d) Digits from 00 to 99 indicate that the wire function has been modified by an "open circuit/closed circuit" circuit element (switches, fuses, relay contacts, etc.). Where a large number of wires are involved, such as transducer leads, and individual amplifier leads to scanning switches, part (d) may also be used to indicate a block or segment of numbers for identifying these wire functions.

(e) Digits from 001 to 999 to identify specific wire functions:

Power functions:		
110 VAC power circuits	1SZ1	A phase
	1SZ2	B phase
	1SZ3	C phase
440 VAC power circuits	1SZ4	A phase
	1SZ5	B phase
	1SZ6	C phase
220 VAC and all other power circuits	1SZ7	A phase
	1SZ8	B phase
	1SZ9	C phase
Synchro functions:		
Low speed stator leads	21SZ1	S1 lead
	21SZ2	S2 lead
	21SZ3	S3 lead
High speed stator leads	21SZ4	S1 lead
	21SZ5	S2 lead
	21SZ6	S3 lead
Any other stator leads	21SZ7	S1 lead
	21SZ8	S2 lead
	21SZ9	S3 lead

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Low numbers for above shields plus "S"	21SZ1S 21SZ4S 21SZ7S	Low shield High shield Other shield
Rotor excitation:		
21SZ10	R1	(This is not confused with the ground notation because the part (b) indicates a synchro function)
21SZZ10	R2	
21SZ10S	Shield for this synchro excitation	

Notes: 1. The shield numbering is added as specified in 3.25.

2. A zero for part (d) will not be necessary except to indicate missing digits: 3SK7 becomes 3SK1007 after passing through a switch indicated by (d). Zeros (0's) indicate the missing digits of part (e) and the first zero of part (d) is omitted. The final complete number would be 3SK01007 with all of the zeros inserted.

3. Use of the part method system of numbering for the wire number with the omission of the zeros, results in a possibility of misinterpretations of the wire number in some cases. But if it is realized that the erroneous interpretations which might be made would be impossible or highly improbable due to the functional use of the wire in the system, no confusion should result. It is more important to keep a log or card file of the wire numbers used in the system so as to avoid duplications.

4. A doubling of the second letter of part (c) shall be used to indicate the second part of a paired circuit such as; the second lead for synchro excitation; the return circuits from each transducer element, amplifier, or scanning switch; and the braid of a coaxial cable. This will obtain better continuity and avoid some overrun of the part (e), 000 to 999, block of wire numbers.

5. A final digit of "0" ordinarily indicates a ground connection, and is to be avoided if possible for use with other wire functions. The basic ground is indicated by "10" (single "0" is not used).

3.26.1 Duplications.- There shall be no duplications of wire numbers except to carry a function through other cables with no change in status after having passed through terminal boards, equipments or junction boxes. The common distribution of information or power and the return circuits shall carry the wire numbers for the functions throughout, provided that there has been no change or modification to the function by switching, relays, amplification, etc. In continuity checking, the wires bearing the same wire numbering should show direct connection indications throughout the sonar system. It is conventional practice to tie all the wires with the same wire numbers together on installation and it is thus important not to duplicate wire numbers for different functions. In making up the installation plan booklet (wire tables) the contractor shall maintain a log or card system of some type showing the wire numbers as assigned in the sonar system arranged numerically under each key number (part (b) of the wire number). A copy of this log shall be presented to the Bureau of Ships, Code 689C, as proof that duplications of wire numbers does not exist in the sonar system. Whatever type of log system the contractor now uses to keep track of the wire numbers he inserts in the tables is acceptable, if it is legible and in sufficient order that wire numbers can be easily located.

3.26.2 Return circuits.- When the return circuit for a function is separate and distinct, use shall be made of the "SKK", "SBB", or "SZZ" (part (c) of the wire number) as the case may be, to indicate this. A common return for several functions will use the wire number for the major or first function assigned, as the wire number for all of the returns. Since a final digit of zero (0) is used to indicate a ground connection, avoid the use of a final digit of zero (0) for any other functional wire number if possible. It is permissible to assign one wire number as system ground, such as "4SK10", and to use this in all instances except where the ground is not common such as a phantom or floating ground.

3.27 Definitions of sonar functions.- Since the list of sonar functions has been extended to include many more terms which are or will be increasingly more prevalent in sonar systems, the following tables have been included to show the derivation of the symbols as well as a list of the most used sonar function symbols:

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Table XI - Basic letter symbols about which the function designation is built.
Table XII - Prefix quantity modifiers to basic symbols - definitions..
Table XIII - Suffix quantity modifiers to basic symbols - definitions.
Table XIV - Numerical modifiers to basic symbols - definitions.
Table XV - Definitions of sonar functions.

Table XV includes the most used sonar functions and indicates how the symbols are derived by proper combinations of the appropriate letters and/or figures from tables XI, XII, XIII and XIV. In this table the sonar or fire control functional symbol is shown followed by the formerly used symbol in parenthesis, if known. Column two contains the short name of the function, if any used, and column three contains the definition of the function as indicated by the symbol.

Table XI - Sonar and fire control function symbols and definitions.

Symbol	Name
A	Angular movement in elevation.
B	Bearing - - - - - - - - - - - - - - -
C	Course - - - - - - - - - - - - - - -
D	Rate of - - - - - - - - - - - - - - -
E	Elevation - - - - - - - - - - - - - -
Ei	Level - - - - - - - - - - - - - - -
F	Missile offset angle - - - - - - - -
G	Gyro angle - - - - - - - - - - - -
H	Linear distance - - - - - - - - -
I	Angle of inclination - - - - - - -
J	Jump deviation - - - - - - - - - -
K	Curvature - - - - - - - - - - - -
L	Sight deflection - - - - - - - - - -
M	Linear movement - - - - - - - -
N	Missile weight - - - - - - - - - -
P	Firing parallax base - - - - - - -
Pn	Navigational parallax base length.
Ps	Position parallax base length.
Q	Quantities related to density - - -
R	Range - - - - - - - - - - - - - - -
S	Lateral angular movement.
T	Time - - - - - - - - - - - - - - -
U	Velocity - - - - - - - - - - - - -
V	Sight angle - - - - - - - - - - - -
W	Wind rate - - - - - - - - - - - - -
Y	Radius of turn - - - - - - - - - -
Z	Cross level - - - - - - - - - - - -

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Table XII - Quantity modifiers to basic symbols.

Alphabetical Prefix	Name	Definition
a	Advance	Portion of quantity measured to advance position.
b	Ballistics	Portion of quantity accounting for superelevation or drift.
c	Computed or generated	Value of a quantity as computed or generated in the mechanism.
d	Designated	Designated value of the quantity.
e	Estimated or error	Estimated value of a quantity or error in that quantity
f	Function	Function of the quantity.
g	Dead time	Correction of the quantity due to dead time.
i	Increment (or decrement)	Increment (or decrement) of the quantity.
j	Computational addition or partial-	Computational addition to the quantity.
jm	Velocity Difference	That portion of the quantity accounting for velocity difference.
jp	Water pressure	That portion of the quantity accounting for change in water pressure.
js	Salinity	That portion of the quantity accounting for change in salinity.
jt	Water temperature	That portion of the quantity accounting for change in water temperature.
jv	Depth difference	That portion of the quantity accounting for change in depth.
l	Initial	Initial value of a quantity.
m	Relative motion	That portion of the quantity accounting for relative motion between own ship and target.
n	Reach	That portion of the quantity accounting for reach.
o	Observed or measured	The observed or measured value of a quantity.
p	Firing parallax	The portion of the quantity accounting for firing parallax.
pn	Navigational parallax	The portion of the quantity accounting for navigational parallax.
ps	Position parallax	The portion of the quantity accounting for position parallax.
q	Corrective input or spot	A corrective input or spot to the quantity.
r	Rate control	Rate control correction to a quantity.
s	Selected	A selected value of the quantity.
u	Initial velocity loss	The portion of the quantity accounting for change in initial velocity.
v	Height or depth	The portion of the quantity accounting for a change in height or depth.
w	Wind	The portion of the quantity accounting for a change in effect of wind.
x	Run difference	The portion of the quantity accounting for run difference.
y	Turning	The portion of the quantity accounting for turning radius.

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Table XIII - Quantity modifiers to basic symbols.

Alphabetical Suffix	Name	Definition
a	Apparent or athwartship	Quantities related to apparent measurement or position.
b	Bearing	Quantities in direction affecting bearing.
c	Weathercocking	A component of weather-cocking wind across the lines of fire and in the wind plane.
d	Deck displacement	Quantities measured in the deck plane.
e	Elevation	Direction affecting elevation.
f	Flight, air	Quantities related to weapon flight through the air.
g	Gun or launcher	Quantities measured with respect to or at the gun or launcher.
h	Horizontal	Quantities measured in the horizontal plane.
i	Inclination	Angular measurement between the horizontal and deck planes.
j	Line of sight to missile	Quantities measured from, to, or about the line of sight to missile.
k	Earth	Quantities expressing earth rates.
m	Missile	Quantities measured from, to, or about line of sight to missile.
o	Own ship	Quantities measured from, to, or about own ship centerline.
p	Past	Quantities relating to past target position.
q	Heading	Compass head of own ship or target.
r	Range	Quantities in direction affecting range.
s	Line of sight	Quantities measured from, to, or about line of sight or director.
t	Target	Quantities measured from, to, or about the target centerline or position.
u	Underwater	Refers to measurements below the horizontal plane.
v	Vertical	Quantities in vertical direction.
w	Wind	Quantities related to wind.
x	East-West	Quantities measured in East-West direction.
y	North-South (true)	Quantities measured from North or in a North-South direction.
'	Prime (before quantity)	Measurement from a normal plane.
'	Prime (after quantity)	Measurement to or in a normal plane.
''	Double prime (before)	Measurement from a plane normal to the slant plane.
''	Double prime (after)	Measurement to or in a plane normal to the slant plane.

* * * * *

Alphabetical Coordinator	Name	Definition
x	Ordinate	An East-West coordinate in the Master System.
y	Abscissa	A North-South coordinate in the Master System.

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Table XIV - Definitions of numbers used in a thrown weapon system.

Numerical Modifier	Name	Definition
0	Will to fire	Quantities measured with respect to will or intent to fire or the optimum time to fire.
1	Firing order	Quantities measured with respect to physical action of firing, i.e., closure of a firing key or any action which initiates an irrevocable firing sequence.
2	Fire	Quantities measured with respect to firing, i.e., ignition.
3	Thrust cut	Quantities measured with respect to the thrust cut-off position.
4	Separation	Quantities measured with respect to separation position.
5	Water entry point	Quantities measured with respect to position where charge enters water.
6	Enabling position	Quantities measured with respect to position where missile is enabled.
7	Future target position	Quantities measured with respect to future target position.
8	Aiming position	Quantities measured with respect to the aiming position.
	USE OF "T"	
T1	Time of firing order	Time of the physical action of firing, i.e., closure of a key or any action which initiates an irretrievable firing sequence.
T2	Time of fire	The time of actual firing, i.e., ignition.
T3	Time to thrust cut-off	The time from instant of firing to instant of thrust cut-off.
T4	Time to separation	The time from instant of firing to instant of rocket and payload separation order.
T5	Time of water entry	The time from instant of firing to instant of water entry.
T6-5	Sinking time	Time from instant charge strikes water to time enabled.
	ETC.	

Table XV - Definitions of sonar function quantities.

Quantity Symbol	Name	Definition
B	Relative target bearing	The angle between the vertical plane through own ship centerline, and the vertical plane through the line of sight, measured in the horizontal plane. Positive angles measured clockwise from own ship centerline.
B5	Relative water entry point bearing	The angle between the vertical plane through own ship centerline, and the vertical plane through the line from own ship to the compound water-entry point, measured in the horizontal plane. Positive angles measured clockwise from own ship centerline.

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Table XV - Definitions of sonar functions quantities (cont'd.).

Quantity Symbol	Name	Definition
sB75	Aim point bearing from future target position	The angle between the vertical plane through the target speed vector, and the vertical plane through the desired missile water-entry point and the center of the target at the future target position. Positive angles measured clockwise from the target speed vector in the horizontal plane.
Ba (Br)	Relative target bearing - - - - - -	Relative target bearing as seen by the observing instrument.
icBa (ΔcBrq)	- - - - - - - - - - - - - - - - - -	Increments of generated sonar range from the Fire Control System (aided tracking).
Bda' (B'r'q)	Sonar train order - - - - - - - - -	The angle between the vertical plane through own ship centerline, and the normal plane through the director and the apparent target position. Positive angles measured clockwise from own ship centerline in the deck plane.
Bdg'	Launcher train order - - - - - - -	Ordered angles between the normal plane through own ship centerline, and the normal plane through the line of fire, measured in the deck plane. Positive angles measured clockwise from own ship centerline.
Bg	Relative launcher bearing - - - -	The angle between the vertical plane through own ship centerline, and the vertical plane through the line of fire, measured in the horizontal plane. Positive angles measured clockwise from own ship centerline.
Bgf''	- - - - - - - - - - - - - - - - - -	The weathercocking angle lying in the wind plane.
Bgy	Launcher bearing - - - - - - - - -	The angle between the North-South vertical plane, and the vertical plane through the line of fire, measured in the horizontal plane. Positive angles measured clockwise from North.
Bm	Relative missile bearing - - - - -	The angle between the vertical plane through own ship centerline and the vertical plane through the line of flight, measured in the horizontal plane. Positive angles measured clockwise from own ship centerline.
Bts (Ta)	Target angle - - - - - - - - - - -	The angle between the vertical plane through the target speed vector, and the vertical plane through the line of sight, measured in the horizontal plane clockwise from the target speed vector.
Bwda	Relative direction of apparent wind	The angle between the vertical plane through own ship centerline, and the vertical plane through the direction from which the apparent wind is blowing, measured in the deck plane. Positive angles measured clockwise from own ship centerline.
Bwy	Direction of true wind - - - - - -	The angle between the North-South vertical plane, and the vertical plane through the direction from which the true wind is blowing, measured clockwise from North.
By	True target bearing - - - - - - - -	The angle between the North-South vertical plane, and the vertical plane through the line of sight, measured in the horizontal plane. Positive angles measured clockwise from North.

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Table XV - Definitions of sonar functions quantities (cont'd.).

Quantity Symbol	Name	Definition
Bya (Bq)	Apparent true target bearing - - -	True target bearing as seen by the sonar equipment.
qBya	- - - - - - - - - - - - - - - - - -	True target bearing correction (manual).
By5	True water-entry point bearing- -	The angle between the North-South vertical plane, and the vertical plane through the line from own ship to the computed water-entry point, measured in the horizontal plane. Positive angles measured clockwise from North.
By8	True bearing to aiming position -	Angle between the North-South vertical plane, and the vertical plane through own ship centerline and the aiming position. Positive angles measured clockwise from North in the horizontal plane. In the absence of wind drift By8=By5.
eBy8	- - - - - - - - - - - - - - - - - -	Error in By8 corresponding to ex and ey.
Co (Co')	Own ship course - - - - - - - - -	The angle between the North-South vertical plane, and the vertical plane through own ship speed vector, measured in the horizontal plane clockwise from North.
Cqo	Own ship heading - - - - - - - - -	The instantaneous ships heading as indicated by the gyro compass, measured in a horizontal plane clockwise from the North-South vertical plane.
Ct (Ct)	Target course - - - - - - - - - - -	The angle between the North-South vertical plane, and the vertical plane through the target speed vector, measured in the horizontal plane clockwise from North.
D	- - - - - - - - - - - - - - - - - -	Differential of real time.
DB (dBq)	Bearing rate - - - - - - - - - - -	The rate of change of sonar bearing.
DBya	Angular - - - - - - - - - - - - -	The rate of change of apparent true target bearing with respect to time.
DEio	Pitch rate - - - - - - - - - - - -	The rate of change of pitch with respect to time.
DMbo	- - - - - - - - - - - - - - - - - -	Own ship velocity across the line of sound.
DMbt	- - - - - - - - - - - - - - - - - -	Target velocity across the line of sound.
DNho (So)	Own ship speed - - - - - - - - - -	The linear rate in the horizontal plane and in the vertical plane through own ship speed vector, due to own ship motion.
DMht (St)	Target speed - - - - - - - - - - -	The linear rate in the horizontal plane and in the vertical plane through the target speed vector due to motion.
DMma	Missile velocity - - - - - - - - -	Accelerometer measurement of missile velocity, measured inside the missile.
DMm3	Missile velocity at thrust cutoff -	Velocity of the missile with respect to the earth's system at thrust cutoff.
DMro	- - - - - - - - - - - - - - - - - -	Own ship velocity along the line of sound.
DMrt	- - - - - - - - - - - - - - - - - -	Target velocity along the line of sound.
DR (dRq)	Sonar range rate - - - - - - - - -	The time rate of change of range.
DRh	Horizontal range rate - - - - - - -	The linear rate in the horizontal plane, and in the vertical plane through the line of sight, due to relative motion, between own ship and target in the frame used by the fire control system.
DZo	Roll rate - - - - - - - - - - - -	The rate of change of roll with respect to time.

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Table XV - Definitions of sonar functions quantities (cont'd.).

Quantity Symbol	Name	Definition
Edua' (E'q')	Apparent line of sight depression	The angle between the deck plane and the line of sight from the transducer to the apparent target position, measured in the normal plane through the line of sight. Positive angles measured downward from the horizontal plane.
Edg'	Launcher elevation order (stabilized)	The ordered angle between the deck plane and the line of fire, measured in the normal plane through the line of fire. Positive angles measured upward from the deck plane.
Ef	- - - - - - - - - - - - - - - - - -	Effective launcher elevation in the presence of weathercocking wind.
Eg	Launcher elevation - - - - - - - -	The angle between the horizontal plane and the line of fire, measured in the vertical plane through the line of fire. Positive angles measured upward from the horizontal plane.
Ego	Launcher elevation - - - - - - - -	Launcher elevation at true launcher bearing Bgy.
Ei	Level angle - - - - - - - - - - - -	The angle between the deck plane and the horizontal plane, measured in the vertical plane through the sonar line of sight. Positive angles measured downward from the horizontal plane on the target side of the ship.
Eio (N)	Pitch - - - - - - - - - - - - - - -	The angle between the deck plane and the horizontal plane, measured in the vertical plane through own ship centerline. Positive angles measured downward from the horizontal plane.
qEio	Roller path pitch correction - - -	The angle between the plane of the roller path and the deck plane, measured in the normal plane through the mount and parallel to own ship centerline. Positive direction is clockwise when viewed inward from own ship starboard side.
Eua (Eq)	Apparent depression angle - - - -	The angle between the horizontal plane and the line of sight to the apparent target position. Positive angles measured downward from the horizontal plane and in the vertical plane through the line of sight.
jstpEua (Eqr)	Refracted depression angle - - - -	The angle between the horizontal plane, which is parallel to the horizontal plane of the axis of the transducer, at the point of refraction of the sound beam and the line of sound, due to (js) salinity, (jt) temperature, or (jp) pressure, measured in the vertical plane through the axis of the refracted sound beam.
icEua (ΔcEq)	- - - - - - - - - - - - - - - - - -	Increments of computed sonar depression/elevation.
qEua (jEq)	- - - - - - - - - - - - - - - - - -	Sonar depression correction (manual).
F	- - - - - - - - - - - - - - - - - -	A unit vector perpendicular to the launcher and in the wind plane.
Hvs (Pvg)	Sonar transducer depth - - - - - -	The depth of the transducer measured in a vertical plane from the surface of the ocean to the horizontal plane of the axis of the transducer.
Mhox	Own ship distance (E-W) - - - - -	The linear motion in the horizontal plane and in the East-West vertical plane due to own ship motion.

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Table XV - Definitions of sonar functions quantities (cont'd.).

Quantity Symbol	Name	Definition
Mhoy	Own ship distance (N-S) - - - - -	The linear motion in the horizontal plane and in the North-South vertical plane due to own ship motion.
P	Launcher parallax base length - -	The total distance from the reference point to the launcher, measured along the launcher parallax base line.
Pdgx	East launcher parallax - - - - - -	The easterly horizontal component of the total launcher parallax.
Pdgy	North launcher parallax - - - - -	The northerly horizontal component of the total launcher parallax.
Pdo	Centerline parallax - - - - - - - -	The component of the total distance from the reference point to the launcher along own ship centerline.
Pdv	Vertical parallax - - - - - - - - -	Height of the launcher trunnoins above the horizontal reference plane, measured in a plane perpendicular to the deck.
PM	- - - - - - - - - - - - - - - - -	Distance from the roll centerline to the trunnion height reference plane (do-dn plane).
PMg	Launcher roll lever arm - - - - -	Length of the line from the roll centerline to the launcher.
PN	- - - - - - - - - - - - - - - - -	Distance from the pitch centerline to the longitudinal parallax reference plane (dn-dv plane).
PNg	Launcher pitch lever arm - - - -	Length of the line from the pitch centerline to the launcher.
Qu	Speed of sound in sea water - - -	(1600 yards per second)
Qv	Air density - - - - - - - - - - -	100% standard density for a standard atmosphere.
Ra (Rq)	Apparent sonar range - - - - - - -	Range to target as measured by the observing instrument along the path of the sound beam.
qRa (jRq)	- - - - - - - - - - - - - - - - -	Sonar range correction (manual).
iRa (ΔcRq)	- - - - - - - - - - - - - - - - -	Increments of sonar range generated in the sonar equipment.
icRa (ΔcRq)	- - - - - - - - - - - - - - - - -	Increments of generated sonar range from Fire Control (aided tracking).
RDB	Linear bearing rate - - - - - - - -	Algebraic sum of own ship and target velocity across the line of sound.
Rh	Computed horizontal range - - - -	The computed projection of present range in the horizontal plane by a vertical plane through the line of sight.
Rha (Rhq)	Apparent horizontal range - - - -	Horizontal range to the apparent target position relative to Ra.
cRha (cRhq)	- - - - - - - - - - - - - - - - -	Computed horizontal sonar range by projection of Ra on horizontal plane.
Rhp	Horizontal range to past position -	The computed projection of past range in the horizontal plane by a vertical plane through the line of sight to the past position.
Rhpx	- - - - - - - - - - - - - - - - -	Easterly component of target past position in the master system.
Rhpy	- - - - - - - - - - - - - - - - -	Northerly component of target past position in the master system.
Rhx	- - - - - - - - - - - - - - - - -	East coordinate of target position in the master system.
Rhy	- - - - - - - - - - - - - - - - -	North coordinate of target position in the master system.

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Table XV - Definitions of sonar functions quantities (cont'd.).

Quantity Symbol	Name	Definition
Rh5	Effective range - - - - - - - - - -	The horizontal range from the reference point to compound water-entry point of the charge at time to fire.
Rh5x	- - - - - - - - - - - - - - - - - -	East coordinate of water-entry point measured in the master system.
Rh5y	- - - - - - - - - - - - - - - - - -	North coordinate of water-entry point, measured in the master system.
Rh8	Horizontal aiming - - - - - - - - -	The horizontal range from own ship to the target, combined with the corrections and predictions necessary to compensate for own ship and target motion during the time of flight, plus ballistic corrections and spots.
eRh8	- - - - - - - - - - - - - - - - - -	Error in Rh8 corresponding to ex and ey.
eRh8x	- - - - - - - - - - - - - - - - - -	East, or x, component of the error between computed water-entry point and desired water-entry point.
eRh8y	- - - - - - - - - - - - - - - - - -	North, or y, component of the error between computed water-entry point and desired water-entry point.
dRm	Designated slant range to missile	Range from the director to the missile along the line of sight, to the missile.
Rp	Slant range to past position - - - -	The distance from own ship to the target's past position, measured along the line of sight to the past position.
Rvua' (Hq)	Apparent target depth - - - - - - -	The depth of the apparent position of the target below the horizontal plane of the transducer, measured in the vertical plane containing apparent target position and own ship.
T4	Time to separation - - - - - - - -	The time from the instant of firing to the instant of rocket and payload separation order.
T5	Time of flight - - - - - - - - - -	The time from the instant of firing to the instant of water-entry.
T6	Enabling time - - - - - - - - - - -	The time from the instant of firing to the instant the missile is enabled (explode in the case of depth charge, or begin search in the case of homing torpedo).
T65	Sinking time - - - - - - - - - - - -	The time from the instant a charge strikes the water to the time that it is enabled, (explode in the case of depth charge, or begin search in the case of a homing torpedo).
Tua	Elapsed time associated with Ra -	One half the total elapsed time from emission of sound wave to return of ping, or 1/2 x 1 (Tu) + Tup.
1(Tu)	- - - - - - - - - - - - - - - - - -	Initial time for emission of the sound beam under water.
Tup	- - - - - - - - - - - - - - - - - -	Time required for sound beam to reach target and return.
Wag	Apparent wind speed at the launcher	The rate of change of the apparent wind motion at the launcher with respect to time. The weathercocking wind vector.
Wagc	Weathercocking cross wind - - - -	The component of weathercocking wind across the line of fire at the launcher.
Wagg	Weathercocking head wind - - - -	The component of weathercocking wind along the line of fire at the launcher.
Wagv	Vertical weathercocking wind - -	The vertical component of weathercocking wind at the launcher.

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Table XV - Definitions of sonar functions quantities (cont'd.).

Quantity Symbol	Name	Definition
Wha	- - - - - - - - - - - - - - - - - -	Apparent wind velocity in the horizontal plane.
Zd	Cross level - - - - - - - - - - - -	Angle between the vertical plane through sonar line of sight, and the normal plane through the sonar line of sight and the deck plane, measured about the axis which is the intersection of the vertical plane through the line of sight and the deck plane. Positive direction is clockwise when viewed along axis inward from target.
Zo	Roll - - - - - - - - - - - - - - - -	Angle between the vertical plane through own ship centerline, and the normal plane through the intersection of the vertical plane through own ship centerline and deck plane, measured about the axis which is the intersection of the vertical plane through own ship centerline and the deck plane. Positive direction is clockwise when viewed inward from own ship bow.

3.28 Assignment of terminal numbers to rotary switches.- All rotary switches shall have terminal numbers assigned in the booklet plan. Wafers are numbered from the top down and these numbers are combined with the terminal letter to form a composite terminal number, that is, 1A, 5H, etc. The top of the switch is defined as the end at which the handle or gear is mounted. The "ON" or most advanced operational mode position of the switch should be considered the normal position of the switch and the orientation established for this position. Clockwise rotation of the handle should turn the switch "ON".

4. QUALITY ASSURANCE PROVISIONS

4.1 This section is not applicable to this specification.

5. PREPARATION FOR DELIVERY

5.1 This section is not applicable to this specification.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the title, number, and the date of this specification.

6.2 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Preparing activity:
Navy - Ships
(Project 5845-N342Sh)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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